



NFL Play-By-Play Models

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NFL Play-by-Play – Context

- **Context:** Whether it be on national tv broadcasts, in coaching rooms, or front office departments, there are many key aspects on the football field that can be predicted based on the understanding of play success, tendencies, and efficiencies. The three models used in this project were predicting run or pass, first down probability, and completion probability on a given play. By combining these insights, decision-makers have a unified framework for evaluating play-by-play efficiency and measuring success rates. Additionally, these models helped determine which pre-snap statistics are most important for predicting play-by-play outcomes.
- **Dataset:** The two datasets used during this process were from NFLFastR. The datasets were from the 2018-2024 seasons.
- **Machine Learning Models Used:** A LightGBM model was used for all three models to capture non-linear relationships in regard to play-by-play outcomes. Each of the three models were evaluated through accuracy, AUC, classification, and Log-Loss metrics. Before training and testing the models, a few of the features were standardized and transformed to better capture situational context so that the model had better predictive performance. Moreover, key parameters were used in each of the models (`n_estimators`, `learning_rate`, and `num_leaves`) to prevent overfitting and balance model complexity.
- **Future Implications:**
 - How can TV and radio broadcasters use these models to provide viewers with real-time insights to enhance their broadcasts?
 - How can NFL front offices use these models to identify what statistics most effectively inform and impact offseason roster decisions?
 - In what ways can coaches use these models to help identify defensive and offensive strategies and tendencies?

NFL Run or Pass Model– Input Variables

- **Target Variables:** Some of the most important target variables used in the Run or Pass model are noted below:

'distance_bucket'	A categorical grouping of the required distance to a first down (short, medium, long, or very long).
'half_seconds_remaining'	Number of seconds left remaining in the half.
'down_distance'	A value that combines down and distance.
'score_differential'	Difference in score between the two teams playing each other.
'is_trailing'	Boolean value representing if the team with possession is trailing or not.
'is_blow_out'	Boolean value representing if the team with possession is losing by more than 14 points.
'shotgun'	Represents whether the offense is lined up in shotgun formation.
'posteam'	Team that possession of the ball.
'is_red_zone'	Represents whether the offense is in the redzone or not.

NFL First Down Probability Model – Input Variables

- **Target Variables:** Some of the most important target variables in the First Down Probability Model are noted below:

'qb_dropback'	Boolean value representing if the QB drops back to pass the ball.
'score_differential'	Difference in score between the two teams playing each other.
'down_distance'	A value that combines down and distance.
'is_trailing'	Boolean value representing if the team with possession is trailing or not.
'is_blownd_out'	Boolean value representing if the team with possession is losing by more than 14 points.
'posteam'	Team that possession of the ball.
'defteam'	Team that is on defense.
'is_red_zone'	Represents whether the offense is in the redzone or not.

NFL Completion Probability Model – Input Variables

- **Target Variables:** Some of the most important target variables in the Competition Probability Model are noted below:

'air_yards'	The distance the football travels from the line of scrimmage to the point of the wide receiver.
'qb_hit'	Boolean value representing if the quarterback was hit by the defense as he is throwing the football.
'qb_scramble'	Boolean value representing if the quarterback intended to throw the ball but instead ran with it past the line of scrimmage.
'pass_location'	Categorical field representing where the ball was thrown by the quarterback (left, middle, or right).
'pass_length'	Categorical classification of how far the football was thrown in the air.
'no_huddle'	Boolean value representing if the offense is running hurry-up offense or not.
'is_red_zone'	Represents whether the offense is in the redzone or not.
'down_distance'	A value that combines down and distance.

NFL Run or Pass Model – Conclusion

Model Goal: The goal of this model is to predict whether a play will be a run or pass. The features in this model were centered around field position, team tendencies, time remaining, and score differential.

Model Used: A LightGBM Classifier Model was used for this situation. This model was used because of its ability to work very quickly with large datasets, address class imbalance, handle high dimensional data, and high predictive accuracy. All these characteristics are highly valuable when working with football play-by-play data. Moreover, the model was split into 80% training and 20% testing datasets. Some of the most important features inputted into this model are defined on slide 3.

Model Evaluation: To evaluate this model, accuracy, recall, precision, and f-1 score were calculated. A classification report was used to see how the model's predictions compared to the actual outcomes. In this model, the overall accuracy was 73% on predicting run or pass. The model also correctly identified 79% of all actual pass plays and 65% of all run plays.

Next Steps: To improve this model in the future, I think it would be great to add offensive and defensive personnel groups/formations. By seeing which players are on the field and the alignment, we would be able to better predict if a play will result in a run or pass. Also, in the future I hope to add rolling averages for run/pass per team and coach into the model.

NFL First Down Probability Model – Conclusion

Model Goal: The goal of this model is to predict the probability of the current play resulting in a first down. The features in this model were centered around field position, team tendencies, time remaining, and score differential.

Model Used: A LightGBM Classifier Model was used for this situation. This model was used because of its ability to work very quickly with large datasets, address class imbalance, handle high dimensional data, and high predictive accuracy. All these characteristics are highly valuable when working with football play-by-play data. Moreover, the model was split into 80% training and 20% testing datasets. Some of the most important features inputted into this model are defined on slide 4.

Model Evaluation: To evaluate this model, one of the most important metrics used was log-loss. Log-loss will penalize overconfident wrong predictions heavily and ensures the model is well-calibrated (rewarding correct predictions). The log-loss for this model was 0.61, indicating that the model is reliable and does a significantly better job calculating first-down probability than randomly guessing. Other evaluation metrics used include AUC, which will reveal how well the model differentiates plays that result in a first down versus those that do not. The AUC was 0.658 for this model (significantly better than guessing).

Next Steps: To improve this model in the future, I think it would be great to add offensive and defensive personnel groups/formations. By seeing which players are on the field and the alignment, we would be able to better predict run or pass. Moreover, including prior play type would be a great way to add value to this model going forward.

NFL Completion Probability Model – Conclusion

Model Goal: The goal of this model is to predict the probability of whether a passing play will result in a completed pass. The features in this model were centered around field position, team tendencies, time remaining, and score differential.

Model Used: A LightGBM Classifier Model was used for this situation. This model was used because of its ability to work very quickly with large datasets, address class imbalance, handle high dimensional data, and high predictive accuracy. All these characteristics are highly valuable when working with football play-by-play data. Moreover, the model was split into 80% training and 20% testing datasets. Some of the most important features inputted into this model are defined on slide 5.

Model Evaluation: To evaluate this model, one of the most important metrics used was log-loss. Log-loss will penalize overconfident wrong predictions heavily and ensures the model is well-calibrated (rewarding correct predictions). The log-loss for this model was 0.56, indicating that the model is reliable and does a better job calculating first-down probability than randomly guessing. Other evaluation metrics used include AUC, which will reveal how well the model differentiates plays that result in a completed pass versus those that do not. The AUC was 0.693 in this model (significantly better than guessing).

Next Steps: To improve this model in the future, I think it would be great to add offensive and defensive personnel groups/formations. Also, adding quarterback and wide receiver mechanic statistics, such as release time, throw depth, route speed, separation at throw, and throw velocity would enhance the model in the future. Quarterback pressure and dropback speed are also great metrics to monitor going forward!